

Experiment 6: React SPA with Routing and API Fetching

App.js

```
import React, { useEffect, useState } from "react";
import {
  BrowserRouter,
  Routes,
  Route,
  Link
} from "react-router-dom";

function Home() {
  return <h2>Home Page</h2>;
}

function Users() {
  const [users, setUsers] = useState([]);
  const [loading, setLoading] = useState(true);

  useEffect(() => {
    fetch("https://jsonplaceholder.typicode.com/users")
      .then((res) => res.json())
      .then((data) => {
        setUsers(data);
        setLoading(false);
      });
  }, []);

  if (loading) return <h3>Loading...</h3>;

  return (
    <div>
      <h2>Users List</h2>
      <ul>
        {users.map((user) => (
          <li key={user.id}>{user.name}</li>
        ))}
      </ul>
    </div>
  );
}

function App() {
  return (
    <BrowserRouter>
      <nav>
        <Link to="/">Home</Link> |
        <Link to="/users">Users</Link>
      </nav>

      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/users" element={<Users />} />
      </Routes>
    </BrowserRouter>
  );
}

export default App;
```

Output

- While fetching: Loading...
- After fetching: Users List containing Leanne Graham, Ervin Howell, Clementine Bauch and other users.

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Aim

1. To develop a React Single Page Application using React Router.
2. To fetch and display user data dynamically from a public API.

Procedure

1. Create a React application.
2. Install React Router DOM.
3. Create Home and Users components.
4. Configure BrowserRouter, Routes and Route.
5. Fetch user data from <https://jsonplaceholder.typicode.com/users>.
6. Store data using `useState()` and fetch it using `useEffect()`.
7. Display Loading... while data is being fetched.
8. Display the users list after retrieval and verify routing.

Sample Input

- Navigate to <http://localhost:3000/users>.
- The application fetches data from <https://jsonplaceholder.typicode.com/users>.

Sample Output

- While fetching: Loading...
- After fetching: Users List containing Leanne Graham, Ervin Howell, Clementine Bauch and other users.

Result

Thus, a React Single Page Application with React Router and API fetching was successfully developed and executed.

[Home | Users](#)

Home Page

[Home | Users](#)

Users List

- Leanne Graham
- Ervin Howell
- Clementine Bauch
- Patricia Lebsack
- Chelsey Dietrich
- Mrs. Dennis Schulist
- Kurtis Weissnat
- Nicholas Runolfsdottir V
- Glenna Reichert
- Clementina DuBuque